



Can't we use any automation tool like terraform or ansible or Catalyst center as a code ? or any other way around ?

Yes — absolutely. There are four mature automation paths to deploy SD-Access, all fully supported by Cisco, and all of them still talk to Catalyst Center as the fabric engine. That part never changes — but the *how you talk to it* is completely flexible. Three ready-to-use files are attached for your POC.

All Roads Lead to Catalyst Center

Every tool — Terraform, Ansible, CC-as-Code, or Python — is just a **different language for calling the same Catalyst Center SDA APIs**. The fabric is always built by CC, the tooling just determines how you express your intent and trigger the deployment.^{[1] [2]}

The Four Paths — Compared

Tool	File Attached	Syntax	Best For	CI/CD Ready
Terraform	sda_terraform_poc.tf	HCL declarative	State management, multi-site rollout, infra teams	✓ Native
Ansible	sda_ansible_poc.yml	YAML playbooks	Network ops teams, existing Ansible AWX/AAP	✓ Native
CC-as-Code	sda_cc_as_code.yml	Native YAML	Simplest syntax, Cisco-native, Git-push deploy	✓ With GitLab/GitHub
Python SDK	sda_poc_deployment.py (prev session)	Python	Custom logic, Meraki Workflow backend, webhooks	✓ With wrapper

Terraform — The Declarative King

Terraform uses the `cisco-en-programmability/catalystcenter` provider on the Terraform Registry. The power here is **state management** — Terraform tracks what's deployed, and running `terraform apply` twice is completely safe (idempotent). The attached `.tf` file handles the full sequence: fabric site → VN → C9500 dual roles → C9300 edge → L3 handoff — all in correct dependency order using `depends_on`.^{[3] [4]}

```
terraform init
terraform plan    # dry-run, shows what will be created
terraform apply   # deploys the fabric
terraform destroy # tears it all down cleanly
```

Cisco Live APJC 2025 demonstrated a full SD-Access fabric site deployed in under 30 minutes using exactly this Terraform provider, including ISE policy and Day-N templating.^[2]

Ansible — Best for Ops Teams

The `cisco.dnac` collection (v6.40.0+) has dedicated modules for every single SDA operation. Key modules for your POC:^[5] ^[6]

- `cisco.dnac.sda_fabric_sites` → Create fabric site
- `cisco.dnac.sda_fabric_devices_workflow_manager` → Assign CP/Border/Edge roles
- `cisco.dnac.sda_host_port_onboarding_workflow_manager` → Port assignment on C9300^[7]
- `cisco.dnac.sda_virtual_networks` → Create CORP_VN

```
ansible-galaxy collection install cisco.dnac
ansible-playbook -i inventory/hosts.yml sda_ansible_poc.yml --ask-vault-pass
```

CC-as-Code — Simplest Syntax of All

This is Cisco DevNet's own Infrastructure-as-Code YAML format for Catalyst Center. It reads exactly like a human-written network design document. The attached `sda_cc_as_code.yml` defines everything — sites, IP pools, fabric, VN, anycast gateways, device roles, and port assignments — in one clean YAML file. It's the highest abstraction level of all four tools and the easiest for someone who doesn't know Terraform HCL or Python.^[8] ^[2]

Which One Should You Use for Your POC?

You want fast, no learning curve	→ CC-as-Code YAML
You have existing Ansible infra	→ Ansible <code>cisco.dnac</code>
You want proper state + multi-site	→ Terraform
You want to wire into Meraki Workflows	→ Python SDK (already built)
You want the full CI/CD pipeline	→ GitLab → Terraform/Ansible → CC → Fabric

The Cisco Live LTRENS-3751 lab (APJC + EMEA 2025) covers the complete CI/CD pipeline: GitLab push → Terraform/Ansible → CC deploys fabric → ISE pushes SGT policy → automated pyATS validation — all in one flow. That is the gold standard and exactly what your POC can evolve toward.^[9] ^[2]

1. <https://developer.cisco.com/docs/catalyst-center/terraform/>
2. <https://www.ciscolive.com/c/dam/r/ciscolive/apic/docs/2025/pdf/LTRENS-3751.pdf>
3. <https://github.com/CiscoDevNet/terraform-provider-catalystcenter>
4. <https://www.ciscolive.com/c/dam/r/ciscolive/emea/docs/2024/pdf/BRKOPS-1183.pdf>
5. https://docs.ansible.com/ansible/latest/collections/cisco/dnac/sda_fabric_sites_module.html
6. https://github.com/DNACENSolutions/dnac_ansible_workflows/blob/main/workflows/sda_fabric_sites_zones/README.md
7. https://docs.ansible.com/projects/ansible/latest/collections/cisco/dnac/sda_host_port_onboarding_workflow_manager_module.html
8. <https://developer.cisco.com/iac/>
9. <https://www.ciscolive.com/c/dam/r/ciscolive/emea/docs/2025/pdf/LTRENS-3751.pdf>
10. https://registry.terraform.io/providers/cisco-en-programmability/catalystcenter/latest/docs/resources/sda_fabric_devices_layer2_handoffs_sda_transits_update
11. <https://www.ciscolive.com/c/dam/r/ciscolive/global-event/docs/2025/pdf/BRKOPS-2142.pdf>
12. <https://developer.cisco.com/docs/catalyst-center/2-3-7-4/terraform/>
13. https://docs.ansible.com/projects/ansible/latest/collections/cisco/dnac/sda_fabric_site_module.html
14. <https://github.com/cisco-en-programmability/catalyst-center-ansible-iac>
15. <https://search.opentofu.org/provider/ciscodcn/catalystcenter/v0.3.3>